

DSC 335 Exam 1

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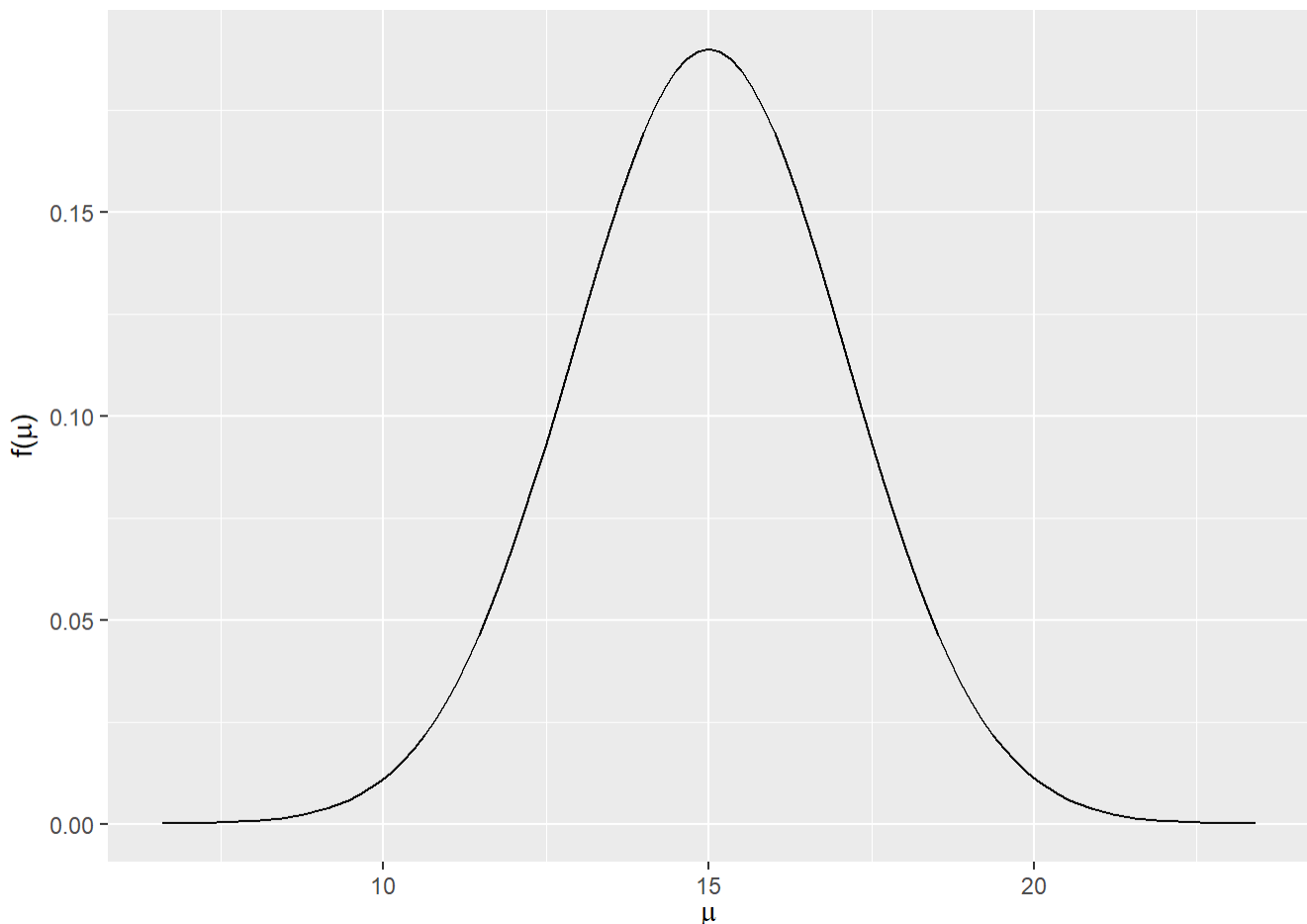
Question 1 (9 points)

Let μ be the average minimum daily temperature in Perth, Australia. You wish to model your prior beliefs about μ with a Normal distribution. You think that 15 degrees Celsius is a reasonable guess at the true value of μ , and you think there is only about a 5% chance that μ would fall outside the interval (11,19). To learn about μ , you plan to use 1000 days worth of temperature data Y , with the assumption that $Y|\mu \sim N(\mu, 7^2)$ and that the values are independent. These data values are included in the variable *mintemp* of the *weather_perth* data frame in the *bayesrules* package.

a. Based on the above information, give a reasonable prior distribution for μ . Justify your choice.

b. Based on your prior in part (a) and the data, give the posterior distribution of μ . Show any code that you run to arrive at your answer in the chunk below.

```
#prior  
plot_normal(15, 2.1)
```



```

# Load required package and data
library(bayesrules)
data(weather_perth)

# Extract the mintemp data (n = 1000)
mintemp_data <- weather_perth$mintemp

# Prior parameters
prior_mean <- 15
prior_sd <- 2.1
prior_var <- prior_sd^2 # prior variance

# Likelihood parameters (known  $\sigma = 7$ )
sigma <- 7
n <- length(mintemp_data) # n = 1000
y_bar <- mean(mintemp_data) # sample mean

# Compute posterior parameters

posterior_var <- prior_var * sigma^2/(sigma^2 + n*prior_var)

posterior_mean <- prior_mean * sigma^2/(sigma^2 + n*prior_var) + y_bar * (n*prior_var^2)/(sigma^2

```

- c. Suppose your friend uses the same prior, but collects only 20 independent observations from a $Y|\mu \sim N(\mu, 7^2)$ distribution. Will their posterior mean be closer to the prior mean or the data mean? Explain your answer. # The posterior mean would be closer to the prior mean because with a relative small sample size of 20, the prior has a greater influence on the posterior model. The sample size might cause a greater numeric value for the posterior variance but the posterior mean shouldn't deviate too much from the prior's.

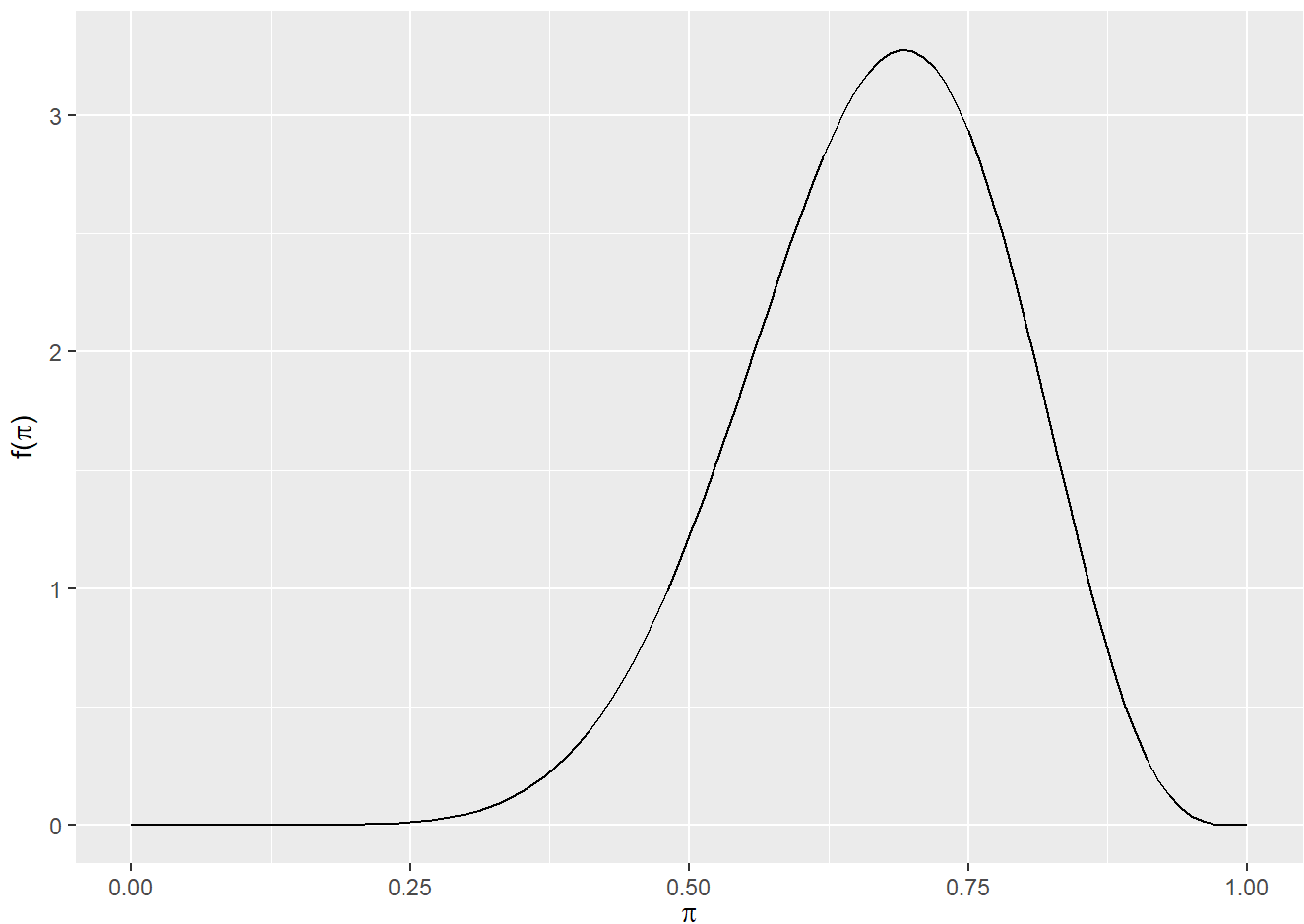
wrong comparison

Question 2 (12 points)

Dr. Daryl thinks the food at Cowan is really good, but he may just have low standards. Let π be the proportion of all Centre students who think the food at Cowan is overrated. We will use a Beta-Binomial model for π .

- a. Suppose we start with a Beta(10,5) prior, which is graphed in the code chunk below. Write a couple of sentences explaining what this distribution tells us about our prior understanding of π , in context.

```
plot_beta(10,5)
```



```
summarize_beta(10,5)
```

	mean	mode	var	sd
1	0.6666667	0.6923077	0.01388889	0.1178511

Prior distribution gives a probabilistic mean of about two-thirds of the population of Centre c

- b. Now assume our class survey is a representative sample of the population of all Centre students. In our sample, 96 out of 156 who responded said they thought Cowan was overrated. Give the posterior distribution of π . # Beta(106, 65)
- c. Suppose Dr. Daryl realizes that he left out 20 survey responses that should have been in the sample above. Describe to Dr. Daryl how he could incorporate the 20 new responses into his model to get a final posterior. Be specific. # Since Bayesian sequential analysis is data order invariant, you could simply assume there's independence among the 20 survey participants and simply repeat the experiment and count the number of successes and failures, before accruing the results into the Bayesian sequential model with this information from the data survey
- d. In the context of this problem, define the term *data order invariance* and tell how it applies to the question above. # Data order invariance indicates that in a Bayesian sequential model, the order in which data collated doesn't matter as long as these experiments are independently done; we

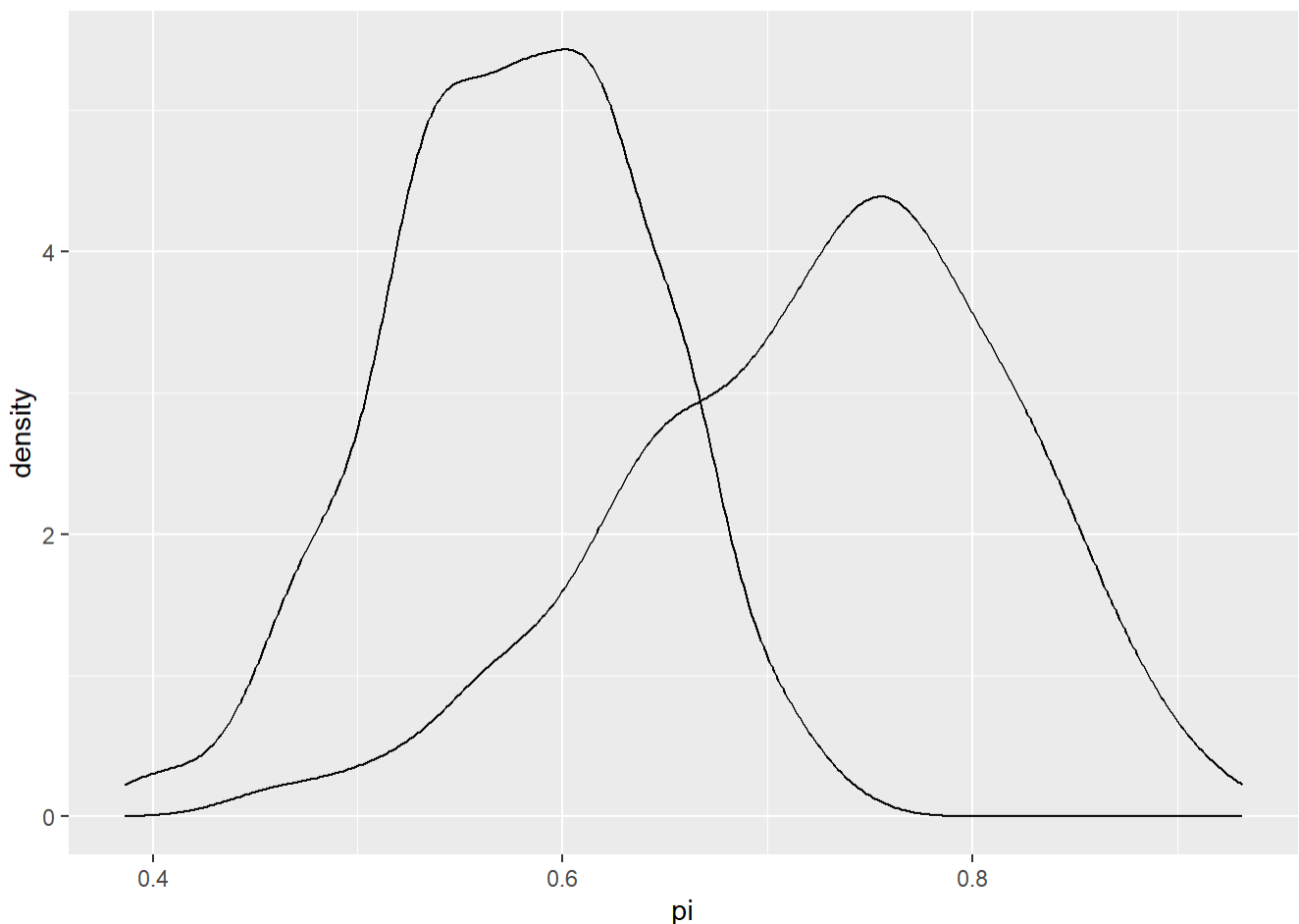
can simply update our posterior model by looking at the kernel involved and adding up the necessary parameters at the very end. ## Question 3 (4 points)

The following code simulates a posterior distribution in two different Beta-Binomial models that have different priors but the same data. Use the code to answer the following questions.

```
#model 1
set.seed(84735)
mod1 = data.frame(pi=rbeta(10000,1,1)) %>%
  mutate(y=rbinom(10000,size=20, prob=pi)) %>%
  filter(y==15)

#model 2
set.seed(84735)
mod2= data.frame(pi=rbeta(10000,20,20)) %>%
  mutate(y=rbinom(10000,size=20, prob=pi)) %>%
  filter(y==15)

ggplot(data=mod1, aes(x=pi))+geom_density() +
  geom_density(data=mod2, aes(x=pi))
```



a. Give the prior distributions of Model 1 and Model 2. Beta(1, 1); Beta(20, 20)

- b. Consider the posterior pdf with the right-most mode (the one around 0.75). Is this the pdf for Model 1 or Model 2? How can you tell? # Model 2 has the right most mode with value of .5. Model 1 is a flat prior which assigns equal plausibility to all probabilistic value $[0,1]$ so the most occurring probability doesn't exist in a Beta(1, 1) model